

Гадиров Наиль
задание 17

1) изменим grub чтобы отобразить меню

```
gadirov@nail-gad:~$ cat /etc/default/grub
# If you change this file, run 'update-grub' afterwards to update
# /boot/grub/grub.cfg.
# For full documentation of the options in this file, see:
#   info -f grub -n 'Simple configuration'

GRUB_DEFAULT=0
GRUB_TIMEOUT_STYLE=menu
GRUB_TIMEOUT=10
GRUB_DISTRIBUTOR=`(. /etc/os-release; echo ${NAME:-Ubuntu} ) 2>/dev/null || echo Ubuntu`
GRUB_CMDLINE_LINUX_DEFAULT="quiet splash"
GRUB_CMDLINE_LINUX=""

# If your computer has multiple operating systems installed, then you
# probably want to run os-prober. However, if your computer is a host
# for guest OSes installed via LVM or raw disk devices, running
# os-prober can cause damage to those guest OSes as it mounts
# filesystems to look for things.
#GRUB_DISABLE_OS_PROBER=false

# Uncomment to enable BadRAM filtering, modify to suit your needs
# This works with Linux (no patch required) and with any kernel that obtains
# the memory map information from GRUB (GNU Mach, kernel of FreeBSD ...)

-- РЕЖИМ ВСТАВКИ --      8,15      Начало
```

2) обновляю

```
gadirov@nail-gad:~$ sudo update-grub
Sourcing file `/etc/default/grub'
Generating grub configuration file ...
Found linux image: /boot/vmlinuz-6.17.0-35-generic
Found initrd image: /boot/initrd.img-6.17.0-35-generic
Found linux image: /boot/vmlinuz-6.17.0-29-generic
Found initrd image: /boot/initrd.img-6.17.0-29-generic
Found linux image: /boot/vmlinuz-6.14.0-34-generic
Found initrd image: /boot/initrd.img-6.14.0-34-generic
Found memtest86+ 64bit EFI image: /boot/memtest86+x64.efi
Warning: os-prober will be executed to detect other bootable partitions.
Its output will be used to detect bootable binaries on them and create new boot entries.
Found Windows Boot Manager on /dev/nvme0n1p1@efi/Microsoft/Boot/bootmgfw.efi
Adding boot menu entry for UEFI Firmware Settings ...
done
gadirov@nail-gad:~$
```

3) КЛОНИМ

```
gadirov@nail-gad:~/prog/kernel$ git clone https://git.kernel.org/pub/scm/linux/kernel/git/torvalds/linux.git
Клонирование в «linux»...
remote: Enumerating objects: 11623749, done.
remote: Counting objects: 100% (4865/4865), done.
remote: Compressing objects: 100% (2119/2119), done.
remote: Total 11623749 (delta 3557), reused 3318 (delta 2728), pack-reused 11618884 (from 1)
Получение объектов: 100% (11623749/11623749), 3.20 ГиБ | 10.49 МиБ/с, готово.
Определение изменений: 100% (9562655/9562655), готово.
Updating files: 100% (94326/94326), готово.
gadirov@nail-gad:~/prog/kernel$
```

4) Новая конфигурация с параметрами defconfig

```
gadirov@nail-gad:~/prog/kernel/linux$ make defconfig
LEX      scripts/kconfig/lexer.lex.c
YACC     scripts/kconfig/parser.tab.[ch]
HOSTCC   scripts/kconfig/lexer.lex.o
HOSTCC   scripts/kconfig/menu.o
HOSTCC   scripts/kconfig/parser.tab.o
HOSTCC   scripts/kconfig/preprocess.o
HOSTCC   scripts/kconfig/symbol.o
HOSTCC   scripts/kconfig/util.o
HOSTLD   scripts/kconfig/conf
*** Default configuration is based on 'x86_64_defconfig'
#
# configuration written to .config
#
gadirov@nail-gad:~/prog/kernel/linux$
```

5) сборка и установка ядра

```
gadirov@nail-gad:~/prog/kernel/linux$ make bzImage && make modules && sudo make modules_install && sudo make install
SYSHDR   arch/x86/include/generated/uapi/asm/unistd_32.h
SYSHDR   arch/x86/include/generated/uapi/asm/unistd_64.h
SYSHDR   arch/x86/include/generated/uapi/asm/unistd_x32.h
SYSTBL   arch/x86/include/generated/asm/syscalls_32.h
SYSHDR   arch/x86/include/generated/asm/unistd_32_ia32.h
SYSHDR   arch/x86/include/generated/asm/unistd_64_x32.h
SYSTBL   arch/x86/include/generated/asm/syscalls_64.h
HOSTCC   arch/x86/tools/relocs_32.o
HOSTCC   arch/x86/tools/relocs_64.o
HOSTCC   arch/x86/tools/relocs_common.o
```

6) обновляю

```
gadirov@nail-gad:~/prog/kernel/linux$ ls -la /boot/vmlinuz-7.1.0-07602-g66affa37cfac
-rw-r--r-- 1 root root 14771200 Jun 17 22:56 /boot/vmlinuz-7.1.0-07602-g66affa37cfac
gadirov@nail-gad:~/prog/kernel/linux$ sudo update-grub
Sourcing file `/etc/default/grub'
Generating grub configuration file ...
Found linux image: /boot/vmlinuz-7.1.0-07602-g66affa37cfac
Found linux image: /boot/vmlinuz-6.17.0-35-generic
Found initrd image: /boot/initrd.img-6.17.0-35-generic
Found linux image: /boot/vmlinuz-6.17.0-29-generic
Found initrd image: /boot/initrd.img-6.17.0-29-generic
Found linux image: /boot/vmlinuz-6.14.0-34-generic
Found initrd image: /boot/initrd.img-6.14.0-34-generic
Found memtest86+ 64bit EFI image: /boot/memtest86+x64.efi
Warning: os-prober will be executed to detect other bootable partitions.
Its output will be used to detect bootable binaries on them and create new boot entries.
Found Windows Boot Manager on /dev/nvme0n1p1@efi/Microsoft/Boot/bootmgfw.efi
Adding boot menu entry for UEFI Firmware Settings ...
done
```

7)

минимум настроек

```
↑(-)
[*] RDC devices
< > RDC R6040 Fast Ethernet Adapter support
[*] Realtek devices
< > RealTek RTL-8139 C+ PCI Fast Ethernet Adapter support
<*> RealTek RTL-8129/8130/8139 PCI Fast Ethernet Adapter support
[*] Use PIO instead of MMIO
[ ] Support for uncommon RTL-8139 rev. K (automatic channel equalization)
[ ] Support for older RTL-8129/8130 boards
[ ] Use older RX-reset method
<*> Realtek 8169/8168/8101/8125 ethernet support
< > Realtek Automotive Switch 9054/9068/9072/9075/9068/9071 PCIe Interface support
[*] Renesas devices
[*] Rocker devices

↑(-)
[*] Broadcom devices
< > Broadcom 43xx wireless support (mac80211 stack)
< > Broadcom 43xx-legacy wireless support (mac80211 stack)
< > Broadcom IEEE802.11n PCIe SoftMAC WLAN driver
< > Broadcom FullMAC WLAN driver
[*] Intel devices
< > Intel PRO/Wireless 2100 Network Connection
< > Intel PRO/Wireless 2200BG and 2915ABG Network Connection
< > Intel Wireless WiFi 4965AGN (iwl4965)
< > Intel PRO/Wireless 3945ABG/BG Network Connection (iwl3945)
<M> Intel Wireless WiFi Next Gen AGN - Wireless-N/Advanced-N/Ultimate-N (iwlwifi)
< > Intel Wireless WiFi DVM Firmware support (NEW)
< > Intel Wireless WiFi MVM Firmware support (NEW)
< > Intel Wireless WiFi MLD Firmware support (NEW)
*** WARNING: iwlwifi is useless without IWL DVM or IWL MVM or IWL MLD ***
```

exit, <?> for Help, </> for Search. Legend: [*] built-in []

--- NTFS file system support

<*> NTFS Read-Write file system support

[] 64 bits per NTFS clusters (NEW)

[] activate support of external compressions lzx/xpress (NEW)

[] NTFS POSIX Access Control Lists (NEW)

8) обновляем grub и перезагружаем

